

Industry Readiness Program

Training Module

Day 1	Core Concepts of OOP in C++ • Introduction to OOP • Difference between Procedural Programming and OOP • Key Concepts: ◦ Class & Object ◦ Encapsulation ◦ Inheritance ◦ Polymorphism ◦ Abstraction • Basic Structure of a C++ Program • Class and Object Syntax
Day 2	Classes and Objects Deep Dive • Defining and Declaring Classes • Access Modifiers: public, private, and protected • Object Instantiation • Scope Resolution Operator (::)
Day 3	Constructors and Destructors • Constructor Types: ◦ Default ◦ Parameterized ◦ Copy Constructor • Constructor Overloading • Destructor Basics • Role of Destructors in Memory Management

Day 4	Encapsulation and Abstraction • Encapsulation (Data Hiding) • Getters and Setters • Abstraction Concepts • Friend Function and Friend Class • Basics of Pointers • this Pointer
Day 5	Inheritance in C++ • Introduction to Inheritance • Types of Inheritance: ◦ Single ◦ Multiple ◦ Multilevel ◦ Hierarchical ◦ Hybrid • protected Access Modifier • Constructor Execution Order
Day 6	Polymorphism in C++ • Compile-time Polymorphism: ◦ Function Overloading ◦ Operator Overloading • Runtime Polymorphism: ◦ Virtual Functions ◦ Abstract Classes ◦ Pure Virtual Functions
Day 7	Dynamic Memory + STL • Dynamic Memory Allocation (new, delete) • Pointers to Objects • Array of Objects using Pointers • Introduction to Templates: ◦ Function Templates

	○ Class Templates ● Introduction to STL: ○ Vectors, Lists, Stacks, Queues ● Iterators and Basic Algorithms
Day 8	File Handling + Mini Project ● File Handling: ○ Reading & Writing Files ● Exception Handling: ○ try, catch, throw ○ Custom Exceptions ● Mini Project: ○ Library Management System ○ CRUD using Classes ○ File Storage Integration ● Final Revision of OOP Concepts
Day 9	Fundamentals of Artificial Intelligence ● What is Artificial Intelligence? ● History and Evolution of AI ● Types of AI: ○ Narrow AI (Weak AI) ○ General AI (Strong AI) ● Characteristics of AI Systems ○ Applications of AI in Real Life: ○ Virtual Assistants ○ Chatbots ○ Smart Devices ○ Gaming ● Advantages and Limitations of AI

Day 10	AI Concepts & Real-World Understanding • AI vs Human Intelligence • Branches of AI: ◦ Natural Language Processing (NLP) ◦ Computer Vision ◦ Robotics ◦ Expert Systems • Decision Making in AI (basic idea) • Automation vs Intelligence • Ethical Issues in AI: ◦ Bias ◦ Privacy ◦ Job Impact • Future Scope of AI
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NEXT IS NOW